Developing a Personal Portfolio Website

From UI Design to Working Website

Grade 11 • Computer Programming (.NET Technology) • 8 hours total

I. General Instructions

You will convert the portfolio website UI that you designed during the previous three-day planning activity into a working website using HTML and CSS only. Use Visual Studio Code to write your code and a web browser to view and test your website. Your final website should reflect your original hand-drawn design as closely as possible.

Learning Objectives

- Create and organize HTML and CSS files in Visual Studio Code.
- Use semantic HTML elements to structure a personal portfolio website.
- Apply CSS properties for colors, typography, spacing, borders, and layout.
- Use CSS Flexbox or Grid to organize website content.
- Create a portfolio website that presents personal information, skills, projects, and contact details.
- Test, debug, and improve a website using a browser.

Required Output

A working personal portfolio website containing:

- Header and navigation menu
- Hero or introduction section
- About Me section
- Skills section
- Projects section with at least three project cards
- Contact section
- Footer

Scope limitation: Use HTML and CSS only. Do not use JavaScript, Bootstrap, or other CSS frameworks. Contact forms and navigation links need only demonstrate the planned layout and basic navigation; actual form submission or backend functionality is not required.

II. Four-Day Activity Schedule

Day	Focus	Main Activities	Daily Output
Day 1	Setting Up the Project and Writing HTML	Create the project folder and files; write the HTML document structure; build the header, navigation, and hero section; preview the page.	Basic HTML homepage
Day 2	Completing HTML and Applying Basic CSS	Complete About Me, Skills, Projects, Contact, and Footer; link CSS; apply colors, fonts, backgrounds, and spacing.	Complete HTML with basic CSS
Day 3	Improving Layout and Visual Design	Use Flexbox or Grid; style project cards, buttons, and navigation; add image placeholders or suitable images; improve responsive behavior.	Styled and organized portfolio

Day 4	Testing, Debugging, and Presentation	Test navigation and visual elements; correct errors; compare with the hand-drawn UI; present the finished website.	Final working portfolio website

III. Detailed Instructions and Activities

Day 1 — Setting Up the Project and Writing HTML

Learning target: Create a valid HTML document and build the initial structure of the portfolio website.

Time allotment: 2 hours (120 minutes)

Time	Activity	Instructions
10 min	Preparation	Open Visual Studio Code and review your hand-drawn UI.
15 min	Project setup	Create a folder named MyPortfolio. Inside it, create index.html and style.css.
20 min	HTML foundation	Write the document declaration, html, head, title, and body elements.
25 min	Header and navigation	Create the header, portfolio name, and navigation links.
30 min	Hero section	Add your name, a short introduction, a role, and a profile-image placeholder.
10 min	Browser preview	Open index.html in a browser and check whether the content appears.
10 min	Saving and checking	Save both files and show the initial output to your teacher.

Activity 1: Create the Project Files

FOLDER STRUCTURE

```
MyPortfolio/
├── index.html
├── style.css
└── images/
```

Save all website files inside the project folder. Place image files in the images folder if you use them.

Activity 2: Write the Initial HTML

Type the following starter code in index.html. Replace the sample name and text with your own information.

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <header>
        <h1>My Portfolio</h1>

        <nav>
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#skills">Skills</a>
            <a href="#projects">Projects</a>
            <a href="#contact">Contact</a>
        </nav>
```

```
    </header>

    <main>
        <section id="home">
            <h2>Hello, I'm Your Name</h2>
            <p>Aspiring Web Developer</p>
            <p>Welcome to my personal portfolio website.</p>
            <a href="#projects">View My Projects</a>
        </section>
    </main>

</body>
</html>
```

Activity 3: Preview the Website

1. Save index.html.
2. Locate the file in your project folder.
3. Double-click it to open it in a web browser.
4. Check that your name, introduction, and navigation appear.
5. Return to Visual Studio Code if you need to correct the content.

Day 1 deliverable: A working HTML page with a header, navigation menu, and introduction section.

Day 2 — Completing HTML and Applying Basic CSS

Learning target: Complete the portfolio content and apply basic styling using an external CSS file.

Time allotment: 2 hours (120 minutes)

Time	Activity	Instructions
10 min	Review	Open yesterday's project and check the HTML structure.
35 min	Content sections	Add About Me, Skills, Projects, Contact, and Footer.
10 min	CSS connection	Verify that style.css is linked correctly.
35 min	Basic styling	Apply colors, fonts, backgrounds, borders, and spacing.
20 min	Browser testing	Refresh the browser, inspect the output, and correct errors.
10 min	Submission checkpoint	Save and show the completed HTML sections and basic styling.

Activity 1: Complete the Remaining HTML Sections

Add the following sections inside the main element, after the Home section and before the closing main tag.

HTML

```
<section id="about">
    <h2>About Me</h2>
    <p>
        I am a Grade 11 Computer Programming student
        interested in web development and technology.
    </p>
</section>

<section id="skills">
    <h2>My Skills</h2>
    <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>Problem Solving</li>
    </ul>
</section>

<section id="projects">
    <h2>My Projects</h2>
```

```
    <article>
        <h3>Project 1</h3>
        <p>Describe your first school project.</p>
    </article>

    <article>
        <h3>Project 2</h3>
        <p>Describe your second school project.</p>
    </article>

    <article>
        <h3>Project 3</h3>
        <p>Describe your third school project.</p>
    </article>
</section>

<section id="contact">
    <h2>Contact Me</h2>
    <p>Email: yourname@example.com</p>
</section>

<footer>
    <p>Copyright &copy; 2026 Your Name</p>
</footer>
```

These sections should be placed within the existing HTML document, not saved as a separate file.

Activity 2: Write the Initial CSS

Open style.css and type the following code.

CSS

```
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f4f4f4;
    color: #333333;
    line-height: 1.6;
}

header {
    background-color: #7a1f36;
    color: white;
    padding: 20px;
    text-align: center;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 10px;
}

section {
    background-color: white;
    margin: 20px;
    padding: 25px;
    border-radius: 8px;
}

h2 {
    color: #7a1f36;
}

article {
    border: 1px solid #cccccc;
```

```
    padding: 15px;
    margin: 10px 0;
    border-radius: 5px;
}

footer {
    background-color: #7a1f36;
    color: white;
    text-align: center;
    padding: 15px;
}
```

Activity 3: Test the CSS

6. Save style.css.
7. Refresh the browser.
8. Check whether the maroon header and footer appear.
9. Confirm that the content sections have backgrounds and spacing.
10. Change the colors to match your original UI design if needed.

Day 2 deliverable: A complete portfolio page with all required content sections and basic CSS styling.

Day 3 — Improving Layout and Visual Design

Learning target: Use CSS layout properties to create an organized and visually consistent portfolio.

Time allotment: 2 hours (120 minutes)

Time	Activity	Instructions
10 min	Design review	Compare the website with the hand-drawn UI.
25 min	Navigation and hero	Improve alignment, spacing, and button appearance.
30 min	Project cards	Arrange project cards using CSS Grid or Flexbox.
20 min	Profile and visual elements	Add a profile image or placeholder and improve typography.
20 min	Responsive design	Make the layout adapt to narrower screens.
15 min	Testing	Preview and correct visual problems.

Activity 1: Improve the Navigation and Hero Layout

Add the following CSS to the end of style.css.

CSS

```
header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 15px;
}

nav {
    display: flex;
    flex-wrap: wrap;
    gap: 10px;
}

nav a {
    margin: 0;
    padding: 8px 12px;
}

nav a:hover {
    background-color: white;
```

```
    color: #7a1f36;
    border-radius: 5px;
}

#home {
    text-align: center;
    padding: 50px 20px;
}

#home a {
    display: inline-block;
    background-color: #7a1f36;
    color: white;
    padding: 10px 20px;
    border-radius: 5px;
    text-decoration: none;
}
```

Activity 2: Arrange the Project Cards

Add the following rule to your CSS. The heading will span the full width above the three cards.

CSS

```
#projects {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 15px;
}

#projects h2 {
    grid-column: 1 / -1;
}

#projects article {
    margin: 0;
    background-color: #ffffff;
}
```

Activity 3: Add Responsive Styling

Add this media query at the end of your CSS file.

CSS

```
@media (max-width: 700px) {
    header {
        flex-direction: column;
    }

    nav {
        justify-content: center;
    }

    #projects {
        grid-template-columns: 1fr;
    }

    section {
        margin: 10px;
        padding: 15px;
    }
}
```

This rule stacks the project cards vertically on narrower screens and adjusts the spacing.

Activity 4: Add a Profile Image (Optional)

If you have an appropriate image, place it in the images folder. Inside the hero section, add:

HTML

```
<img src="images/profile.jpg"
```

```
    alt="Profile photo of Your Name">
```

Replace the filename with your actual image filename. Add CSS to control image size, shape, and spacing. Use a placeholder if you do not want to use a personal photograph.

Activity 5: Visual Inspection

- The navigation links are readable and properly spaced.
- The project cards are aligned.
- The chosen color theme is consistent.
- The website remains readable on a narrow browser window.
- All sections are arranged in the order shown in your original drawing.

Day 3 deliverable: A styled portfolio with organized project cards and responsive layout.

Day 4 — Testing, Debugging, and Presentation

Learning target: Test the website, correct errors, and explain how the HTML and CSS create the intended design.

Time allotment: 2 hours (120 minutes)

Time	Activity	Instructions
15 min	Final code review	Check the HTML structure, file names, and CSS link.
25 min	Navigation testing	Test all section links and the project button.
25 min	Debugging	Correct missing content, broken images, and CSS problems.
20 min	UI comparison	Compare the website against the original hand-drawn design.
25 min	Presentation	Demonstrate the website and explain selected design choices.
10 min	Final submission	Save all files and submit the project folder as instructed.

Activity 1: Test All Navigation Links

Click each navigation link and verify that it leads to the correct section. The links must match the corresponding section IDs.

- href="#about" must lead to id="about".
- href="#skills" must lead to id="skills".
- href="#projects" must lead to id="projects".
- href="#contact" must lead to id="contact".

Activity 2: Debug the Website

- The CSS file is not connected to the HTML file.
- A section ID does not match a navigation link.
- An image does not appear because of an incorrect filename or path.
- A missing closing tag affects the page layout.
- Text or cards overlap or become difficult to read.
- The layout does not adapt properly to a narrow screen.

Activity 3: Present Your Website

Each learner should briefly explain:

11. The purpose of the portfolio.
12. How the HTML structures the page.
13. How CSS controls the colors, spacing, and layout.

14. How the website compares with the original UI drawing.
15. One coding problem encountered and how it was corrected.

Day 4 deliverable: A completed and tested personal portfolio website, ready for teacher evaluation.

IV. Technical Requirements and Submission Guidelines

Requirement	Minimum Expectation
HTML	Valid document structure and semantic elements
CSS	External style.css file
Navigation	Links to all main page sections
Content	Home, About Me, Skills, Projects, Contact, and Footer
Projects	At least 3 project cards
Layout	Consistent alignment, spacing, and typography
Responsive design	Basic adaptation for narrow screens
Code organization	Meaningful file names and readable indentation
Original design	Based on the learner's own hand-drawn UI

Submission Folder

FOLDER STRUCTURE

```
MyPortfolio/
├── index.html
├── style.css
└── images/
    └── profile.jpg
```

The image file is optional. If images are used, include them in the submitted folder. Submit the complete project folder using the method specified by your teacher.

V. Assessment Rubric

Total: 100 points

Criteria	Description	Points
HTML structure and content	Correct page structure and all required sections	25
CSS styling	Appropriate colors, typography, spacing, and visual consistency	20
Layout and responsiveness	Organized project cards, alignment, and narrow-screen adaptation	20
Functionality and testing	Working section links and corrected errors	15
Design implementation	Similarity to the original UI plan and effective design decisions	10
Code organization and presentation	Readable code, proper files, and clear explanation	10
TOTAL		100

VI. Final Checklist for Learners

Check each item before submitting your project:

- ☐ I created index.html and style.css in my project folder.
- ☐ My HTML includes all six required portfolio sections.
- ☐ My CSS file is correctly linked to the HTML file.
- ☐ My portfolio includes at least three project cards.
- ☐ My navigation links lead to the correct sections.
- ☐ My layout uses consistent colors, spacing, and alignment.

- ☐ My website works at a narrower browser width.
- ☐ I checked and corrected coding errors.
- ☐ My final website reflects my original hand-drawn UI.
- ☐ I saved all necessary project files for submission.

Learning progression: Plan the UI → Build the HTML structure → Style with CSS → Test and present.